FLUIDS AND BARRIERS OF THE CNS
REVIEW

The cellular lining of a continuous fluid compartment: an interstitial cell atlas and the Interstitial Hydraulic Gating hypothesis

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Abstract

Background. The interstitium was long treated as inert packing between epithelium, vasculature and parenchyma. Three lines of work have made that position untenable: direct visualization of fluid-filled interstitial spaces in living human tissue, evidence that these spaces are structurally continuous across tissue and organ boundaries, and the demonstration that solute clearance from the brain depends on advective flow through perivascular compartments continuous with them. Each describes a space. None resolves which cells build it, maintain it, or set its mechanical properties.

Scope. This review assembles the cellular lining of that compartment. Thirteen named stromal and interstitial populations, spanning fascia, vasculature, lymphoid tissue, gut and meninges, are described across largely non-overlapping literatures whose communities rarely cite one another, producing marker collisions that propagate into single-cell annotation: fasciacytes, telocytes and adventitial fibroblasts share CD34/PDGFRA profiles; mesenchymal stromal cells overlap cultured fibroblast signatures; and a proposed fourth meningeal layer remains disputed. Populations are admitted here on one criterion, stated explicitly and applied throughout: a cell qualifies if it sets the mechanical or transport properties of a fluid-conducting interstitial compartment. Positive and negative markers, developmental origin, niche and molecular regulation are integrated for each population and for organ-specific derivatives in heart, lung, kidney, gut and skin.

Hypothesis. Applying that criterion yields a testable claim, developed before the atlas rather than after it. The Interstitial Hydraulic Gating hypothesis holds that interstitial volume, set by local matrix protein composition and systemic hydration, gates the patency of the axons, capillaries and perivascular conduits the interstitium bounds, so that compaction and edema are opposite failures of one gate rather than unrelated pathologies. Its central prediction is that regional vulnerability tracks the hydration-sensitivity of local matrix composition rather than the intrinsic properties of neurons or vessels. Competing cell-autonomous and aquaporin-centred accounts are stated alongside it, with the observations that would separate them. An interactive three-dimensional model of the compartment is deposited openly; Figures 3 and 4 are deterministic renderings of it, so the predictions can be reproduced or contradicted directly.

Keywords: interstitium; interstitial fluid; glymphatic system; perivascular space; stromal cell heterogeneity; telocytes; meninges; extracellular matrix; tissue mechanics; interstitial hydraulic gating

Background: a fluid compartment that was not supposed to be there

The interstitium spent most of the twentieth century defined by exclusion: the space that was left once epithelium, vessels and parenchyma had been accounted for. That definition survived partly because the standard preparation destroyed the thing being defined: fixation collapses fluid-filled spaces, so histology returned dense connective tissue where living tissue holds an open, irrigated lattice. The compartment was routinely eliminated before it could be seen.

The interstitium has been directly visualized in living human tissue. In vivo confocal laser endomicroscopy of the human bile duct, and subsequently of dermis, fascia, and perivascular tissue more broadly, revealed a previously unappreciated, fluid-filled interstitial space supported by a lattice of thick collagen bundles and lined by the kind of fibroblast-like cells discussed throughout this review [32]. Subsequent tracer studies using tattoo pigment, colloidal silver, and hyaluronic acid staining showed that these spaces are not confined to single organs: they are structurally continuous from skin into subcutaneous fascia, from colonic submucosa through the muscularis propria into mesenteric fascia, and along the perivascular and perineural sheaths that thread between organs, with an estimated total fluid volume exceeding that of the combined cardiovascular and lymphatic systems [33]. The same continuity has since been demonstrated for the lung, where interstitial spaces link the peribronchovascular interstitium to adjacent mediastinal and pleural fascia [25]. Taken together, these findings reframe the populations catalogued here less as a loose assortment of cell types that happen to share the adjective interstitial, and more as the cellular lining of a single, physically continuous, fluid-filled compartment running through nearly every organ, a framing that strengthens the case for the cross-tissue annotation effort I call for below.

The central nervous system arrived at the same reassessment along a separate route. Iliff and colleagues described cerebrospinal fluid moving through perivascular spaces and clearing interstitial solutes including amyloid β [14]; meningeal lymphatic vessels, distinct from that perivascular pathway, were characterized in mouse dura [15] and then visualized noninvasively in humans [16]; and Møllgård and colleagues proposed a further mesothelial layer subdividing the subarachnoid space [17]. Two decades of work thus converged on a compartment that is continuous, fluid-conducting, and functionally coupled to the tissue around it, from opposite ends of the body and with almost no shared citations.

What none of this work resolves is cellular. The spaces have been imaged, traced, modelled and, in the CNS, tied to clearance and disease. The cells that build the lattice, secrete the matrix that sets its hydraulic properties, line the conduits, and change state when the compartment does have been described separately, by communities studying fascia, lymphoid organs, gut motility, vascular biology and the meninges. These communities rarely read one another. This review assembles that lining, and then asks what follows mechanically from having assembled it.

What unifies these populations

A list of cell types needs an admission rule. "Historically neglected" describes the literature rather than the biology, and would admit any population that happened to be unfashionable. Anatomical adjacency is also insufficient: not all thirteen populations form the interstitium, and several sit in it without building it.

The criterion used here is functional. **A population is included if it sets the mechanical or transport properties of a fluid-conducting interstitial compartment**, by secreting the matrix that determines how far that compartment swells and collapses, by forming its structural lining or lattice, by controlling the caliber of a conduit that runs through it, or by remodelling it in response to injury. The thirteen populations admitted on this basis, and the anatomical sites at which each is best characterized, are mapped in Figure 1. The criterion admits fasciacytes through hyaluronan secretion, pericytes through capillary caliber control, perivascular and meningeal fibroblasts through conduit lining, adventitial fibroblasts as the progenitor pool that remodels the matrix, lymphatic endothelium as the drainage outlet, and interstitial macrophages through matrix turnover and vascular support. Interstitial cells of Cajal and telocytes are included on contested grounds, addressed where each is discussed.

The criterion has three consequences. First, it is falsifiable per population: a cell admitted on it can be excluded by showing its matrix output or structural role does not measurably alter compartment behaviour. Second, it predicts the crosswalk in Figure 2b. If membership is defined by compartment function, populations should map onto compartments many-to-many rather than one-to-one, which is what the marker-transfer failures documented throughout this review look like from the other side. Third, it makes a mechanical claim unavoidable. If these cells are grouped by their effect on a fluid compartment, the question of what that compartment does to its neighbours is not an addendum to the atlas but its reason for existing. That question is taken up next, before the populations themselves, because it determines what is worth recording about them.

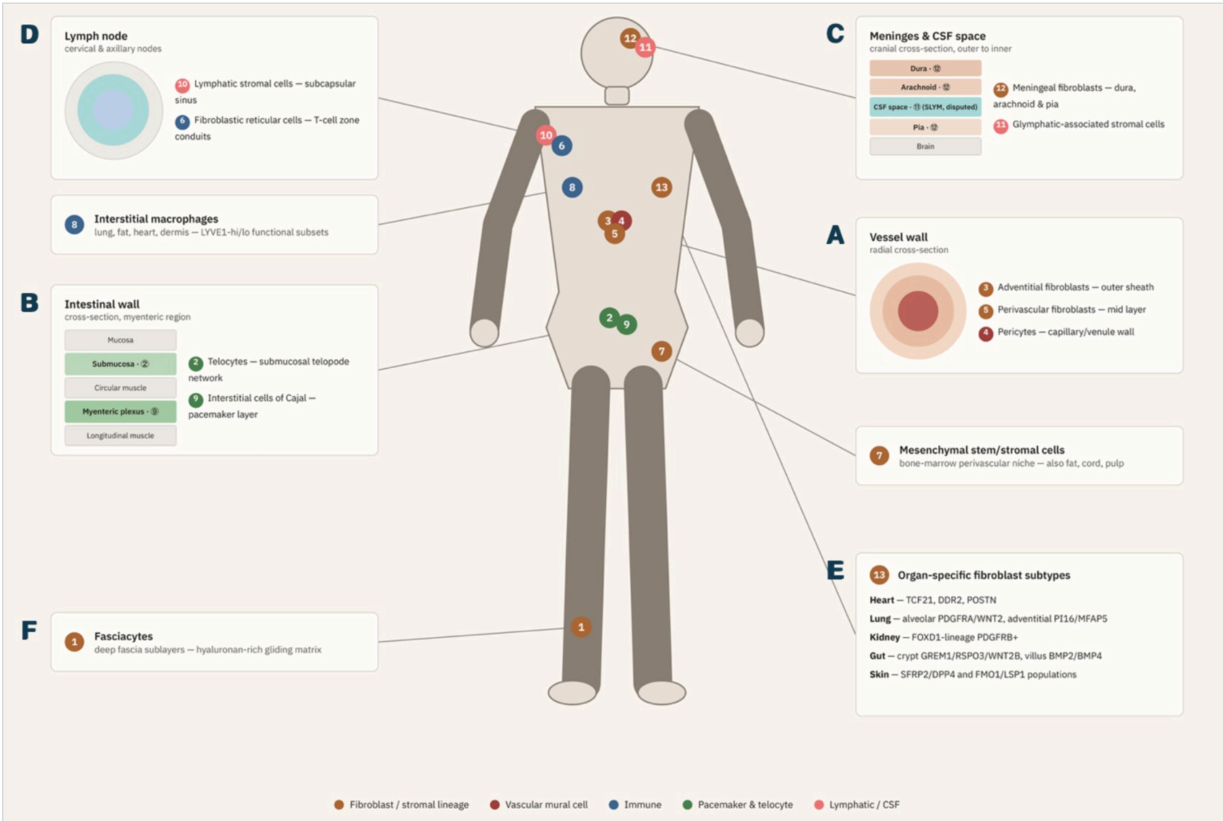

Figure 1. Thirteen interstitial and stromal cell populations mapped onto a schematic human body, with callout numbers positioned at representative anatomical sites and connected by leader lines to detailed inset panels. (A) Vessel wall, radial cross-section: adventitial fibroblasts (3) form the outer sheath, perivascular fibroblasts (5) occupy the mid layer, and pericytes (4) wrap the capillary/venule wall. Mesenchymal stem/stromal cells (7), shown in a separate inset, occupy the adjacent bone-marrow perivascular niche and are also found in fat, cord, and dental pulp. (B) Intestinal wall, myenteric region: telocytes (2) form a submucosal telopode network and interstitial cells of Cajal (9) constitute the pacemaker layer of the myenteric plexus. (C) Meninges and CSF space, cranial cross-section, outer to inner: meningeal fibroblasts (12) define the dura, arachnoid, and pia, while glymphatic-associated stromal cells (11) occupy the perivascular and subarachnoid CSF spaces, including the contested SLYM layer (dashed outline). (D) Lymph node, subcapsular to cortex: lymphatic stromal cells (10) line the subcapsular sinus and fibroblastic reticular cells (6) build the T-cell zone conduit network; interstitial macrophages (8), shown separately at chest level, are distributed as LYVE1-high and LYVE1-low functional subsets across many organs rather than being confined to the lymph node. (E) Organ-specific fibroblast subtypes (13) in heart, lung, kidney, gut, and skin, each specializing the shared adventitial or perivascular stromal template described in the main text. (F) Fasciacytes (1), residing in deep fascia of the limbs and trunk, secrete the hyaluronan-rich matrix that permits gliding between fascial sublayers. Colors indicate the five broad lineage categories defined in the legend strip at the top of the figure.

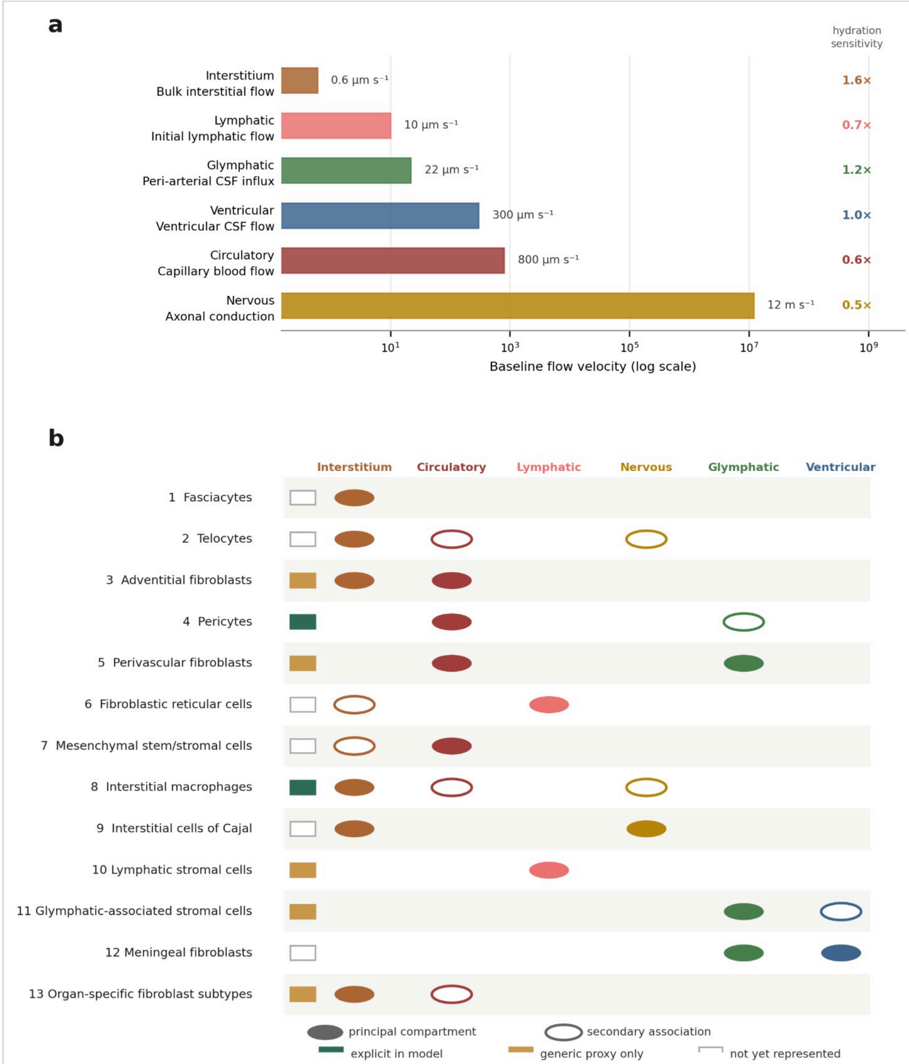

Figure 2. Stromal cell identity mapped onto fluid-transport compartment. (a) Baseline flow velocity across the six fluid and transport compartments that intersect the interstitial space, plotted on a logarithmic scale, with each compartment's relative sensitivity to systemic hydration shown at right. Bulk interstitial flow is simultaneously the slowest compartment, roughly three orders of magnitude below capillary blood flow, and the most hydration-sensitive, so that shifts in systemic water content alter interstitial transport disproportionately relative to vascular transport. Velocities are order-of-magnitude parameterizations drawn from published measurements rather than constants fitted to any single dataset: bulk interstitial flow at 0.1-2 μm/s, measured in normal tissue by fluorescence photobleaching [34]; peri-arterial cerebrospinal fluid flow of order 20 μm/s, measured by particle-tracking velocimetry in mouse pial perivascular spaces [35]; lymph flow of order 10 μm/s in single lymphatic capillaries of human skin [36]; and capillary blood flow of 0.1-1 mm/s [37]. The ventricular cerebrospinal fluid and nervous compartments are placed on the same logarithmic axis by order of magnitude only. No value on this axis should be read as a measured constant for human tissue in vivo; the axis is intended to establish the roughly three-order-of-magnitude separation between interstitial and vascular transport. (b) Crosswalk between the thirteen interstitial and stromal populations catalogued in this review (rows) and the compartments they line (columns). Filled circles denote the principal compartment for a given population; open circles denote a documented secondary association. Note that no compartment is lined by a single population and no population is confined to one compartment, which is the structural reason marker sets developed in one organ or subfield transfer poorly to another. The chip at the left of each row indicates whether that population is currently rendered explicitly in the interactive model described under Resource availability, represented only by a generic fibroblast or endothelial proxy, or not yet represented; this is provided so that readers using the model can see directly where its cellular resolution presently stops.

The Interstitial Hydraulic Gating hypothesis: interstitial volume as a mechanical gate on the systems it bounds

Neurodegeneration and cerebral ischemia are studied as separate problems with separate culprits, misfolded proteins, vascular occlusion, glymphatic and lymphatic failure, yet each unfolds at the same anatomical crossroads: the interstitial compartment that surrounds and physically bounds every axon, capillary, and perivascular conduit in the brain. I therefore propose the Interstitial Hydraulic Gating (IHG) hypothesis (Box 2): that the interstitium is a mechanically active compartment whose protein-determined, hydration-dependent volume gates the patency of these neighboring systems, so that deviation in either direction, toward dehydration (compression, perivascular collapse) or toward edema (raised pressure, rupture, blockage), converges on a single mechanical route to neuronal and vascular injury. Under this framing the separate culprits above are downstream readouts of one upstream variable, and the populations catalogued in this review are the cells that set it.

Three independently established observations sit side by side in the literature without being joined. First, the interstitium is a physically continuous, fluid-filled compartment that runs from skin into fascia and along perineural and perivascular sheaths between organs [32,33]. Second, clearance of interstitial solute from the brain depends on advective movement through perivascular spaces, and a reaction-advection-diffusion model of that route shows that the accumulation of heavy aggregates is suppressed exponentially as advection rises above a modest threshold, Péclet number Pe of order 1, so that aggregates build up once flow falls back toward and below that value [14,20]. Third, capillary caliber and perivascular integrity are under pericyte control, and hereditary disruption of NOTCH3 signaling in that population produces progressive small-vessel disease [7]. Read together, they permit a single upstream variable: the width of the fluid film in the perineural and perivascular interstitium. Because this compartment is the most hydration-sensitive in the body, a systemic water deficit should narrow it before it measurably narrows the vascular compartments, with three simultaneous consequences - degraded ionic buffering in the immediate axonal environment, loss of perivascular advection sufficient to carry the clearance route out of its advection-dominated regime, and increased mechanical load on the pericyte-endothelial unit that maintains capillary caliber. The ordering this predicts is: interstitial narrowing, then impaired clearance and ionic instability, then chronic axonal stress, then the small-vessel and neurodegenerative phenotypes that are currently attributed to the vascular or parenchymal compartments alone.

I state this as a hypothesis rather than a finding because no study cited here measured perineural film width as an independent variable, and the mechanism is therefore inferred rather than demonstrated. It is, however, falsifiable. The framework's central prediction is that regional and temporal vulnerability should track the hydration-sensitivity of local interstitial protein composition, the hyaluronan, proteoglycan and collagen makeup that determines how far a given interstitial film swells and collapses per unit of water shift, rather than the intrinsic properties of neurons or vessels alone. Where two regions carry comparable neuronal and vascular burden, the one whose matrix is the more hydration-labile should fail first; and within a region, the timing of failure should follow systemic water flux more closely than aggregate or occlusion load. Three further predictions follow directly. Graded systemic dehydration should reduce perivascular tracer transport in advance of any detectable change in capillary diameter or perfusion. Clearance deficits induced this way should be reversible on rehydration up to some threshold and irreversible beyond it, which would date the transition from functional to structural injury. And stromal populations that line this compartment - perivascular fibroblasts (5), glymphatic-associated stroma (11), and the meningeal fibroblast layers (12) - should show a coordinated transcriptional response to hydration state, distinct from the injury signature of the adventitial progenitor pool (3), if the compartment is a regulated structure rather than passive space. Figures 3 and 4 state this ordering; they do not test it. The hydration series in Figure 3 shows the parameter that drives the hypothesis, and the scenario presets in Figure 4b-d are visual expressions of it.

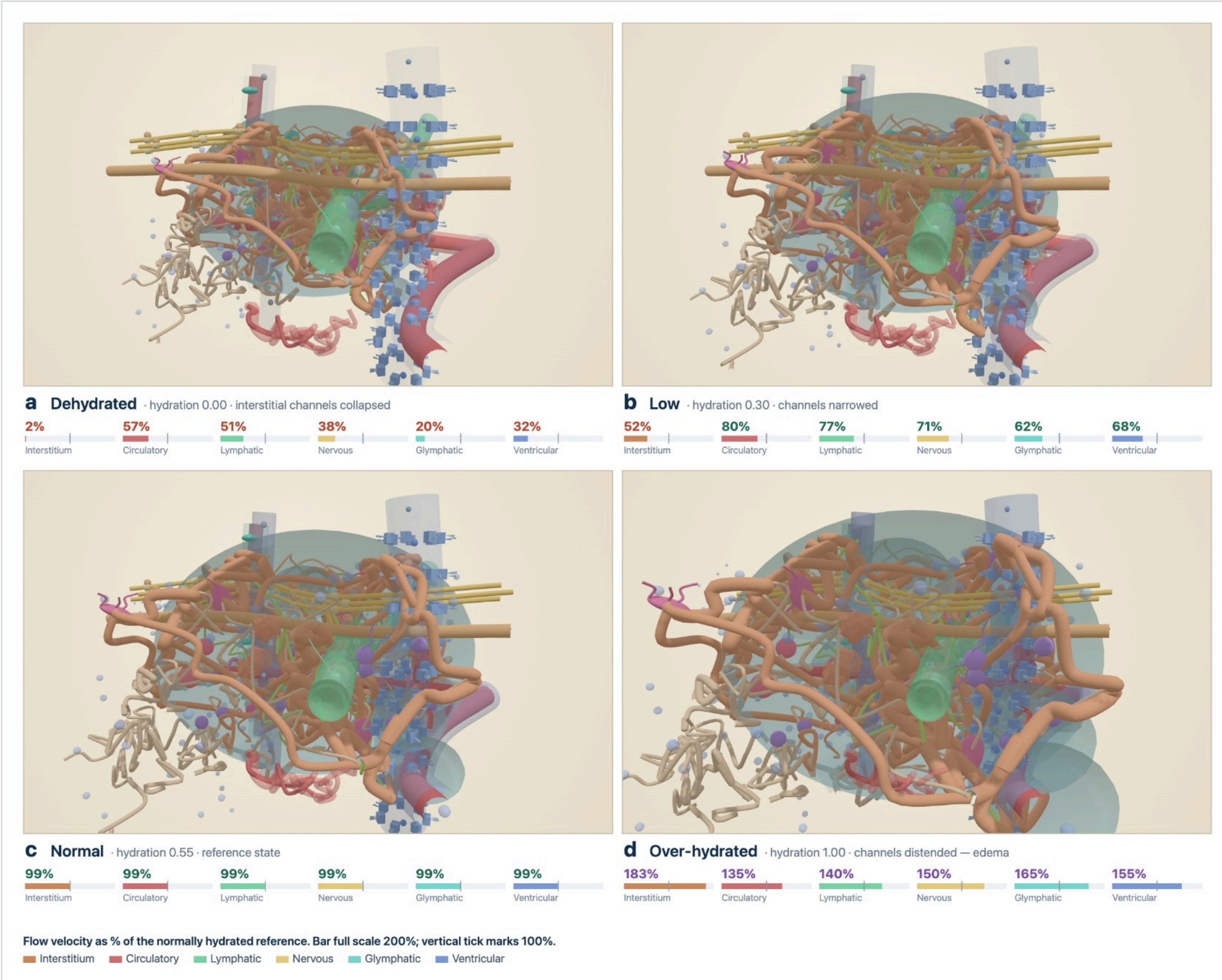

Figure 3. Hydration state deterministically reconfigures every fluid compartment. Four states of the same cortical and meningeal block, rendered from the interactive model at normalized hydration h = 0.00 (a, dehydrated), 0.30 (b, low), 0.55 (c, reference) and 1.00 (d, over-hydrated). Bars beneath each panel give flow velocity in each of the six compartments as a percentage of the normally hydrated reference (full scale 200%, tick mark at 100%). Bulk interstitial flow is the most hydration-sensitive compartment in both directions, falling to 2% of reference when dehydrated and rising to 183% when over-hydrated, while circulatory flow moves only between 57% and 135%; this is the disproportion plotted in Figure 2a, shown here as geometry rather than as a curve. Pre-lymphatic channel width, matrix openness, the interstitial cuffs surrounding vessels and nerves, and the opacity of every fluid-filled space are driven from the single hydration parameter, so the populations that line these compartments - adventitial fibroblasts (3), perivascular fibroblasts (5), and glymphatic-associated stroma (11) - undergo the state change together rather than independently. Percentages are relative to the reference state of the same parameterization and inherit the sources and caveats given in the legend to Figure 2; they are not measured constants.

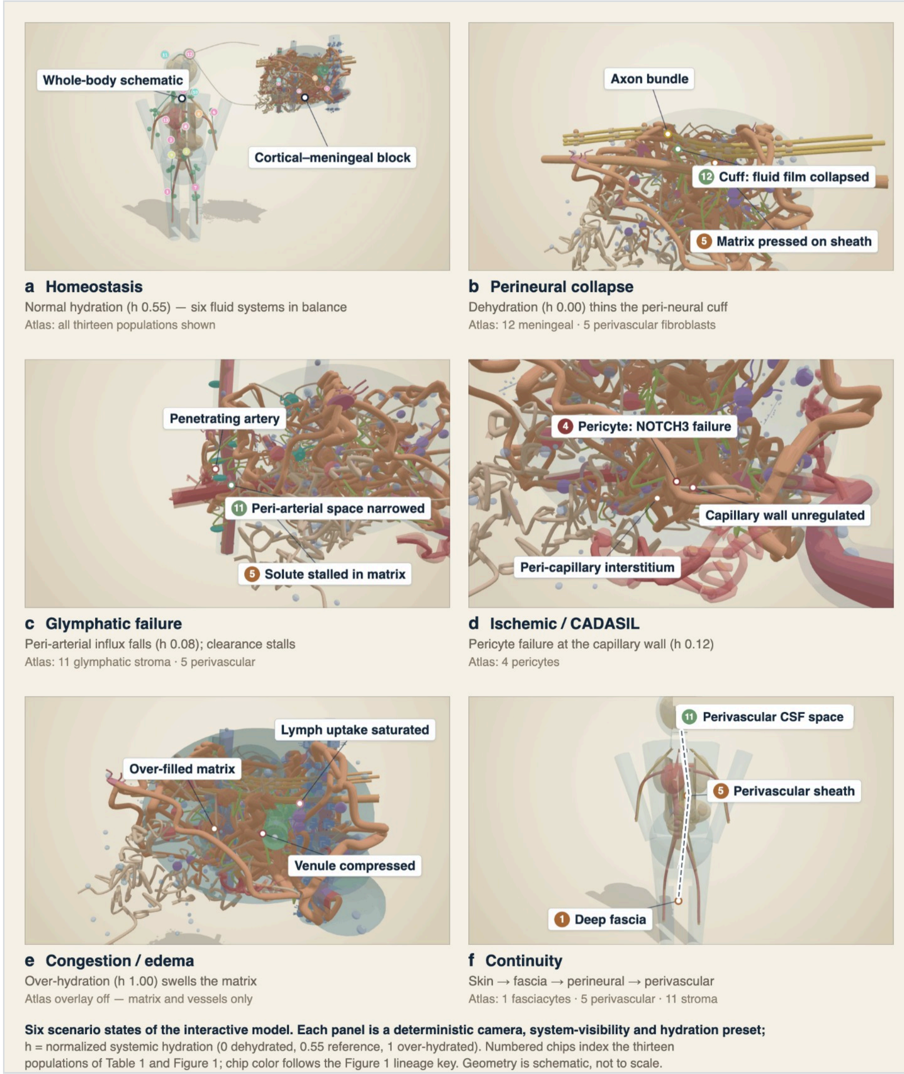

Figure 4. Six scenario states linking cell-atlas populations to compartment pathophysiology. Each panel is a deterministic camera, system-visibility and hydration preset of the same interactive model, captured through the figure-export hook of the archived build, and each is shown at whichever scale carries its point: the whole-body schematic with its cortical and meningeal sampling block called out (a), the microstructural block alone (b-e), or the body alone (f). Labeled chips mark the structures in contact at each conflict point; chip numbers and colors follow the population numbering and lineage key of Figure 1 and Table 1. (a) Homeostasis, h 0.55, with all thirteen populations pinned. (b) Perineural collapse, h 0.00: the axon bundle, the peri-neural fluid film collapsed against it where meningeal fibroblasts (12) line the sheath, and interstitial matrix pressed onto that sheath alongside perivascular fibroblasts (5) - the entry point of the hypothesis set out above. (c) Glymphatic failure, h 0.08: a penetrating artery, its narrowed peri-arterial (AQP4) space occupied by glymphatic-associated stroma (11), and solute stalled in the surrounding matrix (5); a model of this route predicts that aggregate buildup is suppressed exponentially as advection rises above Péclet number of order 1, so the narrowing shown here returns the pathway to a regime in which heavy aggregates accumulate [20]. (d) Ischemic small-vessel disease, h 0.12: a pericyte (4) on the capillary wall - the population in which NOTCH3 disruption causes CADASIL [7] - with loss of caliber control at the wall and the peri-capillary interstitium alongside. (e) Congestion and edema, h 1.00: the over-filled matrix (blue) compresses a neighboring venule as lymphatic uptake saturates; atlas overlay off, so matrix and vessel geometry alone are visible. (f) Continuity: one fluid space running from deep fascia, lined by fasciacytes (1), along the perivascular sheath (5) to the perivascular CSF space (11) [32,33]. Geometry is schematic and not drawn to anatomical scale.

Competing accounts, and what would separate them

Three alternatives account for much of the same ground as Interstitial Hydraulic Gating.

Clearance failure is molecular, not mechanical. On this account, impaired solute clearance reflects loss of aquaporin-4 polarization at the astrocytic endfoot and altered transporter expression, with perivascular geometry a downstream consequence rather than the driving variable. It predicts that clearance deficits track AQP4 localization and astrocytic state, and that they can be produced with interstitial geometry held constant, which IHG cannot accommodate.

Small-vessel disease is mural-cell-autonomous. NOTCH3 disruption damages pericytes directly, and the resulting loss of capillary caliber control is sufficient to explain progressive white-matter injury without an interstitial term. It predicts that vessel pathology precedes any interstitial change, and that matrix composition is irrelevant once mural cell genotype is known.

Interstitial change may be a consequence rather than a cause. Compartment narrowing in disease may be a downstream readout of parenchymal loss, inflammatory matrix remodelling, or altered perfusion: real, measurable, and without causal weight.

These are separable by experiment. The discriminating variable is ordering under a controlled hydration challenge, because only IHG predicts that interstitial geometry moves first. Graded systemic dehydration should reduce perivascular tracer transport before any detectable change in capillary diameter, perfusion or AQP4 distribution; the molecular account predicts the reverse order, and the consequence account predicts no interstitial change at all in the absence of parenchymal injury. Second, matrix composition should carry predictive weight independent of vascular genotype: among regions matched for neuronal and vascular burden, the one with the more hydration-labile matrix should fail first, which the cell-autonomous account excludes by construction. Third, reversibility on rehydration should have a threshold, recoverable while the change is geometric and fixed once structural, whereas neither alternative predicts a hydration-defined transition point.

None of these experiments has been done, so the hypothesis remains untested. The discriminating observations are set out here so that it can be.

Assembling the lining: thirteen populations

Each population below is assessed against the criterion above: what it contributes to compartment mechanics, and how securely its identity rests on marker and lineage data. These populations are summarized in Table 1, with full marker sets in Table 2. Where a population's identity is disputed, the dispute is stated rather than resolved.

At least thirteen named interstitial and stromal populations now appear across separate literatures: connective-tissue populations described mainly in fascia and sensory-organ research (fasciacytes, telocytes), vascular-associated populations (pericytes, adventitial and perivascular fibroblasts), lymphoid stromal populations (fibroblastic reticular cells, lymphatic stromal cells), a functionally defined progenitor population (mesenchymal stem or stromal cells), immune cells embedded within the interstitium (interstitial macrophages), a specialized gut pacemaker population (interstitial cells of Cajal), meningeal and glymphatic-associated stroma, and organ-specific fibroblast subtypes in heart, lung, kidney, gut, and skin. These populations and their representative anatomical sites are mapped in Figure 1, an overview summarized further in Table 1, with the full marker gene set for each population given in Table 2.

These populations are rarely considered together, because the communities that study them, fascia anatomists, cardiac and gastrointestinal physiologists, lymphoid-organ immunologists, and neuroscientists studying the meninges, rarely cite one another. This creates two practical problems. First, nomenclature collisions, most seriously between telocytes and interstitial cells of Cajal [3,4], risk misannotation of single-cell clusters across tissues, since a computational pipeline trained on one organ's marker set will silently mislabel an analogous but molecularly distinct population in another. Second, a fragmented marker literature makes it difficult to build the kind of cross-tissue reference atlas that has already proven valuable for classical fibroblasts and mural cells, and that is now standard practice for immune cell annotation. The consequences reach past nomenclature: several of these populations are directly implicated in disease, from kidney and lung fibrosis to gut motility disorders and meningeal biology relevant to neurodegeneration, so a marker gene misassigned in a discovery dataset can propagate into a mischaracterized therapeutic target.

Populations are described below in terms of discovery, tissue niche, developmental origin and molecular regulation, drawing on cross-tissue single-cell RNA sequencing atlases published within the past several years and cross-checked against each population's original description.

Fasciacytes and telocytes are recently named, morphologically defined connective-tissue cells

Fasciacytes

Fasciacytes (Box 2) were described by Stecco and colleagues [5] as round to oval, hyaluronan-secreting cells lining the internal sublayers of deep fascia, distinct from the elongated classical fibroblast and functionally analogous to synoviocytes and ocular hyalocytes. Positive markers are limited to S100A4, HAS2, and vimentin, with CD68 negativity excluding a macrophage identity. Their developmental origin has not been separately fate-mapped and is presumed to be local mesenchyme, most likely of fibroblast lineage, though this has not been formally tested with lineage tracing. No transcriptomic cluster has yet been confidently assigned to this population; it remains defined by histology and immunohistochemistry rather than single-cell sequencing.

Telocytes

Telocytes (Box 2), named by Popescu and Faussone-Pellegrini [3] after earlier work under the name interstitial Cajal-like cells, are defined less by a marker gene than by morphology: a small cell body extending one to five long **telopodes** (Box 2) that alternate thin podomers with dilated podoms. They have been reported in nearly every organ, typically marked by CD34 and PDGFRA, with tissue-variable KIT and PDGFRB. Whether telocytes constitute a distinct lineage or represent CD34-positive adventitial or perivascular fibroblasts imaged by electron microscopy remains an open dispute, since CD34/PDGFRA positivity alone will not separate them from adventitial fibroblasts in dissociated single-cell data, a limitation that should temper confidence in any telocyte-labeled cluster drawn from scRNA-seq alone. A recent single-cell transcriptomic survey of human dorsal root ganglia used combined CD34 and PDGFRA expression to define a telocyte population and proposed thousands of candidate ligand-receptor contacts with adjacent sensory neurons, though this interactome remains to be functionally validated, and no transcription factor or epigenetic program specific to telocyte identity has yet been defined in any organ. Whether the telopode network constitutes a genuine structural scaffold stabilizing stromal-neuronal contacts, comparable in spirit to the membrane-associated periodic skeleton recently shown to stabilize neuron-epidermal attachment in *Caenorhabditis elegans* [6], is untested; the invertebrate scaffold is a spectrin-based cortical lattice rather than a telopode network, so the analogy is functional rather than one of shared molecular identity, and whether it is applicable to vertebrate telocyte biology at all remains to be tested.

A universal adventitial and perivascular fibroblast scaffold underlies most organs

Cross-tissue single-cell atlases spanning seventeen mouse and human organs identified two conserved, universal fibroblast states present in nearly all tissues [1]: a PI16- and DPT-positive adventitial state occupying vessel adventitia and organ capsules, and a second, COL15A1-marked state occupying a more perivascular position. Lineage tracing supports the adventitial state as a progenitor reservoir giving rise to more specialized fibroblasts during homeostasis and to activated fibroblasts during injury. A parallel cross-tissue analysis [2] established criteria for discriminating fibroblasts from mural cells, reinforcing PDGFRA and CD34 as core adventitial markers alongside PI16, MFAP5, and mouse Ly6a (Sca1), with secreted collagen I/III, decorin, and fibrillin.

A structurally distinct perivascular fibroblast population, described in the same brain vasculature atlas that defined pericytes [7], occupies the space just outside the pericyte and basement-membrane layer on vessels larger than capillaries, marked by COL1A1, COL1A2, COL3A1, DCN, and ANPEP (CD13), and distinguished from pericytes by absence of RGS5. In the central nervous system this population lines the Virchow-Robin perivascular space (Box 2), a site directly relevant to the glymphatic-associated stroma discussed below. Because both adventitial and perivascular fibroblasts sit immediately adjacent to pericytes and mural cells without a validated positive marker that excludes the other, distinguishing the two in practice typically requires combining several markers, PDGFRA level, RGS5 status, and anatomical position, rather than any single gene.

Pericytes are the most rigorously defined mural cell population

Pericytes, first observed by Rouget in the 1870s, received a modern molecular definition from single-cell transcriptomics of the mouse brain vasculature [7], which distinguished them from vascular smooth muscle cells and from perivascular fibroblasts using PDGFRB, RGS5, CSPG4, ANPEP, KCNJ8, and ABCC9, together with the capillary-specific markers HIGD1B, ART3, and VTN. PDGFB-PDGFRB signaling from endothelium is the central recruitment axis, with ANGPT1-TIE2 and NOTCH3-JAG1 contributing; FOXC2, NOTCH3, and ZEB1 have been implicated transcriptionally. NOTCH3 mutations that disrupt this signaling cause CADASIL (Box 2), a hereditary small-vessel disease. Developmental origin is mesodermal in most organs, with a well-documented neural crest contribution to craniofacial and forebrain vasculature, a distinction that likely underlies some of the regional heterogeneity observed in pericyte transcriptomes across the brain. Because this population rests on large, replicated datasets and a clean surface panel (MCAM/CD146, PDGFRB, CSPG4), it is a useful benchmark against which the specificity of markers proposed for the more contested populations below should be judged.

Fibroblastic reticular cells organize the conduit architecture of lymphoid tissue

Fibroblastic reticular cells form the three-dimensional conduit network of lymph node and splenic T-cell zones, channeling lymph-borne antigen through collagen bundles they themselves ensheathe. Droplet-based single-cell sequencing of mouse lymph node [8] resolved nine non-endothelial stromal clusters, including CCL19-high and CCL19-low T-zone reticular cells, marginal reticular cells, follicular dendritic cells, and a CD34-positive capsule and medullary adventitia population. The standard discriminating gate is PDPN-positive, CD31-negative, CD45-negative. CCL19 and CCL21A recruit T cells and dendritic cells via CCR7, IL7 supports T-cell survival, and RANK-RANKL signaling from lymphoid tissue inducer cells drives differentiation, though no single transcription factor defines the lineage. Lymphoid stromal biology has produced some of the field's most granular single-cell atlases, yet these remain almost entirely uncited in the fascia, cardiac, and lung fibroblast literatures describing structurally and functionally analogous immune-organizing stroma. Fibroblastic reticular cells derive from mesenchyme responsive to lymphoid tissue organizer signaling during lymph node development. Disruption of this network during chronic viral infection or checkpoint-blockade immunotherapy alters T-cell access to antigen.

Mesenchymal stem or stromal cells are a functional definition, not yet a stable molecular identity

The population traces to Friedenstein's identification of clonogenic fibroblast colony-forming units in bone marrow [9], formalized by the International Society for Cellular Therapy's minimal criteria [10]: positivity for NT5E, THY1, and ENG (CD73/CD90/CD105), negativity for PTPRC, CD34, ITGAM/CD14, CD79A/CD19, and HLA-DR, plus clonogenicity and tri-lineage differentiation. RUNX2, PPARG, and SOX9 govern osteogenic, adipogenic, and chondrogenic branch fates rather than baseline identity. Functionally similar populations are isolated from bone marrow, adipose stromal vascular fraction, umbilical cord Wharton's jelly, and dental pulp, with mesodermal origin in most cases and a documented neural crest contribution for some craniofacial populations. Bivalent chromatin at the osteogenic, adipogenic, and chondrogenic lineage gene sets, poised for activation, has been described as a feature of the undifferentiated state, alongside progressive DNA methylation changes at these loci across passage-associated senescence in culture, one of the few populations in this review with an explicit epigenetic signature attached to its identity.

Critically, in single-cell transcriptomic space, cultured mesenchymal stromal cells substantially overlap with the adventitial and perivascular fibroblast signatures described above. Whether they constitute a molecularly distinct in vivo population, or a functional state accessible to several fibroblast subtypes once removed from their niche and expanded in culture, remains unresolved (Box 1).

Interstitial macrophages divide into a perivascular and a nerve-associated subset

The molecular dichotomy within interstitial macrophages was defined by fate-mapping and single-cell sequencing across murine lung, fat, heart, and dermis, with confirmation in human tissue [11]. LYVE1-high, MHC-II-low, CX3CR1-low macrophages localize adjacent to blood vessels, support vascular integrity, and restrain fibrosis; LYVE1-low, MHC-II-high, CX3CR1-high macrophages localize adjacent to nerve fibers and are more antigen-presenting. The core surface gate is CD45-positive, FCGR1 (CD64)-positive, MERTK-positive. Ontogeny is mixed: yolk-sac-derived embryonic macrophages self-renew locally alongside continuous monocyte replenishment, in a ratio that varies by organ and age. Tissue-specific enhancer landscapes are established when local niche signals act on PU.1 (SPI1)-primed chromatin [12], giving interstitial macrophage identity an explicit epigenetic mechanism that most of the other populations discussed here still lack. The perivascular, LYVE1-high subset is of particular translational interest because its depletion worsens experimental lung fibrosis.

Interstitial cells of Cajal generate gut pacemaker activity and must not be confused with telocytes

Interstitial cells of Cajal (Box 2), named for Ramon y Cajal's original silver-stain descriptions, were functionally confirmed as gastrointestinal pacemaker cells when Huizinga and colleagues [4] showed that the W/kit gene is required both for the cells themselves and for intestinal slow-wave activity. They occupy the myenteric plexus, intramuscular positions, and, in the small intestine, the deep muscular plexus, and depend on KIT signaling from KITLG (stem cell factor) for development and maintenance. KIT (CD117) remains the primary sorting marker, ANO1 (TMEM16A) encodes the pacemaker chloride channel, and ETV1 drives the pacemaker-specific transcriptional program. The nomenclature risk here is substantial: Popescu's group originally called telocytes 'interstitial Cajal-like cells' [3], and the two populations are still occasionally conflated in downstream literature despite being molecularly and functionally distinct, KIT-positive pacemaker cells versus CD34/PDGFRA-associated stromal cells. Loss of KIT signaling depletes interstitial cells of Cajal and is implicated in diabetic gastroparesis; no comparable disease mechanism has been established for gut telocytes.

Lymphatic stromal cells and a contested glymphatic-associated layer

Lymphatic stromal cells

Lymphatic endothelial identity was established through Prox1 knockout studies in mouse [13], which showed that Prox1 drives lymphatic vessel formation from venous endothelium. PROX1, PDPN, LYVE1, and FLT4 (VEGFR3) together define lymphatic endothelium, with PROX1 the critical discriminator from blood endothelium given that both share PECAM1. VEGFC-FLT4 signaling drives lymphangiogenesis, and SOX18 and NR2F2 (COUP-TFII) act alongside PROX1 in fate commitment. The surrounding perilymphatic stromal sheath, mesenchymal in origin, forms the subcapsular sinus-lining stroma within lymph nodes and the perilymphatic sheath of peripheral lymphatic vessels more broadly, though this stromal component is considerably less well characterized at the single-cell level than the lymphatic endothelium it surrounds.

A contested glymphatic-associated layer

The **glymphatic system** (Box 2), describing cerebrospinal fluid movement through perivascular spaces, was introduced by Iliff and colleagues [14]; meningeal lymphatic vessels, distinct from this perivascular pathway, were rediscovered in mouse dura [15] and confirmed in humans noninvasively [16]. In 2023, a further stromal layer was proposed, the **SLYM** (Box 2), or subarachnoid lymphatic-like membrane, described as a Prox1-positive mesothelium subdividing the subarachnoid space [17]. Despite the name, its reported phenotype is PDPN- and PROX1-positive but LYVE1- and VEGFR3-negative, which does not support a lymphatic identity in the conventional sense, and subsequent rebuttals [18,19] argue the Prox1-positive cells are embedded within the arachnoid barrier layer rather than forming an independent membrane. Independent of this anatomical dispute, biophysical modeling of the perivascular clearance pathway itself has quantified how effectively it can remove aggregation-prone proteins: a reaction-advection-diffusion model of amyloid-β clearance through the glymphatic route found that clearance of heavy aggregates falls off exponentially once advective flow exceeds a modest Péclet number threshold [20], underscoring that the functional stakes of this stromal compartment do not depend on resolving the SLYM controversy. Prox1 positivity alone therefore cannot be used to annotate a SLYM cluster in dissociated meningeal data.

Meningeal fibroblasts define three molecularly distinct layers

Layer-specific meningeal identities were established by single-cell sequencing of the developing mouse meninges combined with a Foxc1 knockout in which meningeal fibroblast development fails [21]. The dura, the outer, vascularized layer that in the adult also carries the meningeal lymphatics discussed above, is marked by CRABP2, MGP, and FXYD5. The arachnoid fibroblast layer is marked by ALDH1A2 (RALDH2) and CRABP2, while a functionally distinct arachnoid barrier-cell sublayer forms epithelial-like tight junctions marked by CLDN11 and E-cadherin. The pia, a single fibroblast layer apposed to the glia limitans, is marked by S100A6, NGFR, and LAMA1/LAMA2. FOXC1 and FOXC2 are required upstream for meningeal fibroblast specification generally, and cranial meninges are predominantly neural-crest-derived rostrally, with a mesodermal contribution caudally and in the spinal meninges. The dura, arachnoid and pia differ substantially in vascularization, barrier function and disease involvement: meningioma, subdural hematoma and CNS lymphatic dysfunction each implicate a different one of these three fibroblast populations rather than a generic meningeal stroma.

Organ-specific fibroblast subtypes specialize a shared stromal template

The populations above provide a shared template that individual organs specialize further. In heart, TCF21 marks the resident cardiac fibroblast lineage, DDR2 is a broadly used surface marker, POSTN is induced upon activation, and WT1 marks the epicardium-derived subset [22]. In lung, the Human Lung Cell Atlas [23] resolved alveolar fibroblasts (PDGFRA, WNT2), adventitial fibroblasts (PI16, MFAP5), and peribronchial fibroblasts (ACTA2); a subsequent spatial multi-omic study [24] identified a rare CCL19/CCL21/CXCL13/FDCSP/GREM1-expressing fibroblast subset at sites of bronchial immune-cell infiltration, functionally reminiscent of the fibroblastic reticular cells described above, and raising the possibility that lymphoid-organizer-like fibroblasts are not restricted to lymphoid organs. These cell-level findings sit within a lung interstitium now shown to be structurally continuous with the mediastinal and pleural fascia beyond it, echoing the pattern established for skin, gut, and liver [25]. The transition from a quiescent TCF21-positive state to an activated, POSTN-expressing, matrix-depositing state is a central event in post-infarction remodeling and a candidate target for antifibrotic intervention.

In kidney, Foxd1-lineage tracing established that nephrogenic stromal progenitors give rise to both PDGFRB-positive pericytes and interstitial fibroblasts, and that this lineage dominates myofibroblast formation in fibrosis [26]; paired single-cell and spatial data have since resolved the specific myofibroblast origins involved in human kidney fibrosis [27]. In gut, a BMP/WNT signaling gradient organizes subepithelial fibroblasts along the crypt-to-villus axis, with GREM1, RSPO3, and WNT2B marking crypt-bottom, stem-cell-supportive fibroblasts and BMP2/BMP4 marking villus-tip fibroblasts [28], while a separate single-cell study mapped structural remodeling of this compartment in inflammatory bowel disease [29]. In skin, two major dermal fibroblast populations, one marked by SFRP2 and DPP4 and the other by FMO1 and LSP1 [30], have proven difficult to map cleanly onto the older papillary/reticular anatomical distinction; a separate dermal papilla population associated with hair follicles is commonly linked to WNT5A, BMP4, and CORIN expression, and lineage tracing has established that dermal fibroblasts derive from at least two developmentally distinct lineages with different fibrogenic potential [31]; the two lineages differ in their propensity to generate fibrotic versus regenerative matrix during wound healing. Across all five organs the pattern is the same: a shared adventitial or perivascular fibroblast template is locally re-specified by organ-specific developmental signals, rather than each organ inventing its stromal compartment independently.

Table 1. Overview of anatomical location, marker genes, and functional role for the thirteen interstitial and stromal cell populations discussed in this review.

#	Cell type	Anatomical location	Marker genes	Purpose
1	Fasciacytes	Deep fascia sublayers (limb, trunk)	S100A4, HAS2, VIM; CD68 negative	Secretes hyaluronan-rich matrix enabling gliding between fascial sublayers
2	Telocytes	Perivascular / subepithelial, most organs	CD34, PDGFRA, KIT/PDGFRB (organ-dependent)	Putative stromal signaling network via telopode contacts; identity vs. adventitial fibroblasts unresolved
3	Adventitial fibroblasts	Vessel adventitia and organ capsules	PI16, DPT, PDGFRA, CD34	Quiescent progenitor pool for specialized and disease-activated fibroblasts
4	Pericytes	Capillary/venule basement membrane, all vascularized tissue	PDGFRB, RGS5, CSPG4, NOTCH3, MCAM	Stabilize and regulate the microvascular wall
5	Perivascular fibroblasts	e.g., cerebral Virchow-Robin spaces	COL1A1/3A1, DCN, LUM, ANPEP	Structural sheath; lines perivascular CSF-influx conduits
6	Fibroblastic reticular cells	T-cell zones of lymph node and spleen	PDPN, CCL19, CCL21A, IL7	Channel antigen and guide T-cell trafficking through a collagen conduit network
7	Mesenchymal stem/stromal cells	Bone-marrow perivascular niche (also fat, cord, pulp)	NT5E/CD73, THY1/CD90, ENG/CD105	Clonogenic progenitor with tri-lineage (bone, fat, cartilage) potential
8	Interstitial macrophages	Non-epithelial compartment of nearly every organ	CSF1R, FCGR1, MERTK, LYVE1/MRC1	LYVE1-hi subset supports vessels; LYVE1-lo subset skews antigen-presenting
9	Interstitial cells of Cajal	Myenteric and intramuscular GI plexuses	KIT/CD117, ANO1, ETV1	Pacemaker cells generating GI slow-wave electrical rhythm
10	Lymphatic stromal cells	Lymphatic vessel wall and lymph-node subcapsular sinus	PROX1, PDPN, LYVE1, FLT4	Line lymphatics; drive lymphangiogenesis via VEGFC-FLT4 signaling
11	Glymphatic-associated stromal cells	Perivascular CSF spaces; subarachnoid (disputed 4th layer)	COL1A1/ANPEP; contested SLYM: PDPN/PROX1/CRABP2	Implicated in CSF influx and clearance of interstitial solutes
12	Meningeal fibroblasts	Dura, arachnoid and pia surrounding the CNS	Dura: CRABP2, MGP. Arachnoid: ALDH1A2. Pia: S100A6, NGFR	Layer-specific structural and barrier support for the CNS
13	Organ-specific fibroblast subtypes	Heart, lung, kidney, gut, skin	TCF21; PDGFRA/WNT2; FOXD1; GREM1/RSPO3; SFRP2/DPP4	Tissue-tailored ECM/signaling niches, e.g. the gut BMP-WNT crypt-to-villus gradient

Row numbers are shaded by lineage, matching Figure 1: ***fibroblast/stromal lineage (1, 3, 5, 7, 12, 13)*** , ***vascular mural cell (4)*** , ***immune (6, 8)*** , ***pacemaker & telocyte (2, 9)*** , ***lymphatic/CSF (10, 11)*** .

Table 2. Detailed positive, negative, surface, secreted, and transcription-factor marker genes for each population, including organ-specific fibroblast subtypes.

Cell type	Positive and enriched genes	Discriminating negative genes	Surface proteins for FACS	Secreted ECM and signaling genes	TF
Fasciacytes	S100A4, HAS2, VIM	CD68	Not established as a validated panel; VIM intracellular with CD68 exclusion is the working gate	HAS2 product hyaluronan	Not characterized
Telocytes	CD34, PDGFRA, KIT, PDGFRB (organ dependent), VIM, CAV1, NES (kidney)	NG2/CSPG4 and PDGFRB negative in renal telocytes specifically; DCN low specificity	CD34, PDGFRA double positive gate, KIT/CD117 in some organs	DCN (weak), telopode caveolae component CAV1	Not well defined
Adventitial fibroblasts	PI16, DPT, PDGFRA, CD34, LY6A/SCA1 (mouse), MFAP5, SCARA5	Low ACTA2, low PDGFRB relative to pericytes	PDGFRA, CD34, SCA1 (mouse)	COL1A1, COL3A1, DCN, FBN1, DPT	PPARG (quiescent state), EGR1
Pericytes	PDGFRB, RGS5, CSPG4, ANPEP, KCNJ8, ABCC9, DES, NOTCH3, MCAM, HIGD1B, VTN, ART3	PECAM1, PTPRC, CLDN5	MCAM, PDGFRB, CSPG4, ANPEP	COL4A1, LAMB1, FN1	FOXC2, NOTCH3, ZEB1
Perivascular fibroblasts	COL1A1, COL1A2, COL3A1, PDGFRA (low), LUM, DCN, ANPEP	RGS5, CLDN5, high PDGFRB relative to pericyte	ANPEP, low PDGFRB	COL1A1, COL3A1, FN1, LUM	Not well defined
Fibroblastic reticular cells	PDPN, PDGFRB, CCL19, CCL21A, IL7	PECAM1, PTPRC	PDPN (gp38), PDGFRB	COL1A1, COL4A1 conduit fibers	No single master TF; RANK to RANKL dependent
Mesenchymal stem or stromal cells	NT5E, THY1, ENG	PTPRC, CD34, ITGAM, CD14, CD79A, CD19, HLA-DR	NT5E (CD73), THY1 (CD90), ENG (CD105)	COL1A1, FN1	RUNX2 (osteogenic), PPARG (adipogenic), SOX9 (chondrogenic)
Interstitial macrophages	CSF1R, FCGR1, MERTK, LYVE1, MRC1, FOLR2, CX3CR1, ADGRE1 (mouse F4/80)	Subset dependent; alveolar markers SIGLEC1, SIGLEC-F excluded	FCGR1 (CD64), MERTK, MRC1 (CD206), LYVE1	MMP9, MMP12, IL10	SPI1, MAFB, MAF
Interstitial cells of Cajal	KIT, ANO1, ETV1	CD34 negative in most gastrointestinal subtypes	KIT (CD117)	KITLG to KIT signaling axis	ETV1
Lymphatic stromal cells	PROX1, PDPN, LYVE1, FLT4	Shares PECAM1 with blood endothelium; PROX1 is the discriminator	PDPN, FLT4 (VEGFR3), PECAM1 (shared)	VEGFC to FLT4 axis, ANGPT2 to TIE1	PROX1, SOX18, NR2F2
Glymphatic associated stromal cells	Perivascular: COL1A1, PDGFRB (low), ANPEP. SLYM: PDPN, PROX1, CRABP2	SLYM is LYVE1, FLT4, CLDN11 and CDH1 negative	PDPN, ANPEP	COL4A1, LAMA1/2 basement membranes; VEGFC to FLT4	PROX1
Meningeal fibroblasts, dura	CRABP2, MGP, FXYD5, COL1A1, COL1A2	ALDH1A2, S100A6, NGFR absent	Not well validated for FACS	COL1A1, COL1A2	FOXC1, FOXC2
Meningeal fibroblasts, arachnoid	ALDH1A2, CRABP2 (fibroblast layer); CLDN11, CDH1 (barrier cells)	S100A6 and NGFR absent	CLDN11 (barrier cells)	Basement membrane collagens, limited data	FOXC1, FOXC2
Meningeal fibroblasts, pia	S100A6, NGFR, LAMA1, LAMA2	ALDH1A2 and CRABP2 absent	NGFR (p75NTR)	LAMA1, LAMA2 pial basement membrane	FOXC1, FOXC2
Organ specific, cardiac fibroblasts	TCF21, DDR2, POSTN (activated), COL1A1, WT1 (epicardial lineage)	Low ACTA2 in the resting state	DDR2	COL1A1, COL3A1, POSTN	TCF21, WT1
Organ specific, lung fibroblasts	Alveolar: PDGFRA, WNT2. Adventitial: PI16, MFAP5. Peribronchial: ACTA2. Immune recruiting: CCL19, CCL21, CXCL13, FDCSP, GREM1	Adventitial and alveolar markers are reciprocally low	PDGFRA	COL1A1, ELN (peribronchial)	Not well defined beyond WNT2 pathway
Organ specific, kidney fibroblasts	PDGFRB, FOXD1 lineage descendants, EPO (erythropoietin producing subset)	Not well defined	PDGFRB, NT5E (CD73)	COL1A1, COL4A1	FOXD1

Organ specific, gut fibroblasts	Crypt bottom: GREM1, RSPO3, WNT2B. Villus tip: BMP2, BMP4	Reciprocal exclusion along the crypt to villus axis	Not well established	COL6A1	No single master TF; BMP and WNT gradient driven
Organ specific, skin fibroblasts	SFRP2, DPP4 (one major population); FMO1, LSP1 (second major population); dermal papilla: WNT5A, BMP4, CORIN	Reciprocal exclusion between the two major populations	DPP4 (CD26)	COL6A5, COL1A1	No single master TF
Organ specific, fascia (classical fibroblasts)	COL1A1, COL3A1, DCN	CD68	Not well established	COL1A1; hyaluronan via neighboring HAS2 positive fasciacytes	Not characterized

Abbreviations: ECM, extracellular matrix; FACS, fluorescence-activated cell sorting; TF, transcription factor.

Conclusions

Individual populations are increasingly well resolved by single-cell transcriptomics within their own subfields, yet almost no cross-referencing occurs between them (Figure 1; Tables 1 and 2). This has concrete costs. A researcher annotating CD34/PDGFRA-positive cells in fascia cannot currently tell from marker genes alone, whether they are looking at telocytes, adventitial fibroblasts, or a novel population, because the relevant reference atlases were built in isolation from one another. The same is true, in different ways, for the boundary between mesenchymal stromal cells and the fibroblast subtypes that apparently give rise to them in culture, and for the still-unsettled anatomical status of the SLYM layer in the meninges (Box 1). Each disagreement turns on a substantive question: whether a given marker combination reflects a stable developmental lineage, a transient activation state, or an artifact of tissue dissociation. The answer differs by population.

The compartment is also mechanically active, which is what Interstitial Hydraulic Gating turns into a testable claim: if interstitial volume gates the systems it bounds, the stromal populations catalogued here are the effectors of that gate, and their matrix output, not only their identity, is the phenotype worth measuring.

I see three priorities for the field. First, cross-tissue atlases in the style of Buechler and colleagues [1] should be extended explicitly to fascia, telocyte-rich organs, and the meninges, where comparable efforts are currently missing, ideally using consistent dissociation and sequencing protocols so that marker comparisons across studies are not confounded by technical batch effects. Second, functional lineage-tracing studies, of the kind already applied successfully to kidney stroma [26], are needed to resolve whether mesenchymal stromal cell identity is niche-intrinsic or culture-induced. Third, the field would benefit from a shared minimal marker panel, analogous to the ISCT criteria for mesenchymal stromal cells [10], that different subfields could adopt when annotating stromal single-cell data, reducing the nomenclature collisions documented throughout this review. None of these priorities requires new technology. They require the connective-tissue research community to read across its own subfield boundaries.

Box 1. Open problems for a unified interstitial cell atlas

- Do telocytes represent a transcriptionally distinct lineage, or are they CD34-positive adventitial or perivascular fibroblasts viewed under electron microscopy?
- Is mesenchymal stromal cell identity niche-intrinsic, or a culture-induced state accessible to multiple fibroblast subtypes once removed from their native environment?
- What is the true anatomical status of the subarachnoid lymphatic-like membrane (SLYM): an independent fourth meningeal layer, or a subcompartment of the arachnoid barrier?
- Can a shared minimal marker panel reliably distinguish fasciacytes, telocytes, and adventitial fibroblasts within a single dissociated tissue sample?

- Do lymphoid-organizer-like fibroblasts found outside lymphoid organs, such as the immune-recruiting subset described in lung, share a developmental program with fibroblastic reticular cells?
- What developmental signals establish the sharp molecular boundaries between dural, arachnoid, and pial fibroblasts, and are these boundaries actively maintained or fixed early in development?

Box 2. Glossary

CADASIL: A hereditary small-vessel disease caused by NOTCH3 mutations that disrupt pericyte-endothelial signaling, leading to progressive white-matter injury and stroke.

FACS (fluorescence-activated cell sorting): A technique that separates cells by surface marker expression, used throughout this review to define candidate isolation panels.

Fasciacyte: A hyaluronan-secreting cell lining internal sublayers of deep fascia that facilitates gliding between adjacent fascial planes.

Glymphatic system: A proposed pathway of cerebrospinal fluid movement through perivascular spaces that clears interstitial solutes, including amyloid β, from brain tissue.

Interstitial cell of Cajal (ICC): A KIT-dependent pacemaker cell of the gastrointestinal tract that generates the electrical slow wave underlying gut motility.

Interstitial Hydraulic Gating (IHG): The hypothesis proposed in this review that interstitial volume, set by local matrix protein composition and systemic hydration, gates the patency of the axonal, capillary, and perivascular compartments the interstitium bounds, so that both dehydration and edema converge on a shared mechanical route to neuronal and vascular injury.

Lineage tracing: A genetic method that permanently labels a cell population and its descendants, used to establish developmental origin, for example of Foxd1-positive kidney stroma.

scRNA-seq (single-cell RNA sequencing): A method that measures gene expression in individual cells; the principal technology underlying the cross-tissue atlases discussed in this review.

SLYM (subarachnoid lymphatic-like membrane): A recently proposed, Prox1-positive mesothelial layer subdividing the subarachnoid space, whose status as an independent fourth meningeal layer is disputed.

Telopode: The extremely long, thin cellular process characteristic of telocytes, alternating narrow podomer segments with dilated podom segments.

Virchow-Robin space: The perivascular space surrounding penetrating cerebral vessels, lined by perivascular fibroblasts and implicated in cerebrospinal fluid movement.

Availability of data and materials

The interactive three-dimensional model of the interstitial compartment used to generate Figures 3 and 4 is openly available as a self-contained HTML application requiring no installation, archived with a permanent identifier: **The Human Interstitium, Interactive 3D Map**, https://doi.org/10.5281/zenodo.21584351 (concept DOI; cite the version DOI corresponding to this manuscript). Source, figure-export hooks and the deterministic scenario presets are included in the deposit, together with the four figures at publication resolution.

Rendering is deterministic: the scenario and hydration settings named in the legends to Figures 3 and 4 reproduce the published panels exactly, and the hydration parameter can be swept continuously across its range. Readers can therefore reproduce, extend or contradict the predictions in this review without re-implementing the model.

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Competing interests

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Claude (Anthropic) to assist with literature search and verification, drafting and organizing text across multiple rounds of revision, and formatting the manuscript and figures to journal specifications. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article, including independent verification of the scientific claims and citations herein.

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